Ahtsham Mustafa Awan

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About Myself

Strong foundation in Geology, Landscape Dynamics, GIS and Remote Sensing concepts and principles, as well as experience with various software tools and programming Languages (Spatial Data Science and Web GIS Development)

Education

MS **National University of Sciences and Technology, Islamabad, Pakistan** — Sept 2021 – Sept 2023
Remote Sensing and GIS

- **Coursework:** Advanced Geographic Information Systems, Advanced Geodatabase and Programming, Advanced Remote Sensing and Image Processing, Natural Hazards and Disaster Management, GIS for Agriculture and Natural Resources, Spatial Data Modelling, Web GIS
- **Thesis:** Comparison of Machine Learning Classifiers to Assess Landscape Dynamics and Their Relationship with Land Surface Temperature, Topography, and Socioeconomic Variables

BS **The University of Azad Jammu and Kashmir, Pakistan** — Sept 2016 – Feb 2021
Applied Geology

- **Major Coursework:** Geomorphology, Economic Geology, Geochemistry, Gemology, Geology of Pakistan, Mineralogy, Petrology, Field Geology, Structural Geology, Geophysics, Rock Mechanics, Geostatistics, Mathematics, Geohydrology, Engineering Geology, Geotectonics, Sedimentation, Environmental Geology, Environmental Hazards, Well Logging, Stratigraphy, General Geology, Surveying and Drawing, Remote Sensing and GIS
- **Thesis:** : Landslide Mapping and Spatial Distribution Analysis from Muzaffarabad to Luat, Neelum Valley, Azad Kashmir, Pakistan

Skills

Software: ArcGIS, ArcGIS Pro, GeoDa, eCognition, Erdas Imagine, QGIS, HEC-RAS, RAMMS, MapTiler, SAS Planet, Latex, Microsoft Office, Adobe Illustrator, Photoshop, After Effects, Premiere Pro
Web Technologies: HTML, CSS (Bootstrap), JavaScript (React, Next.js, OpenLayers, Leaflet, ArcGIS JS API, Google Maps API, Cesium.js, Mapbox GL, GEE JavaScript API), PostgreSQL, Microsoft SQL, Git
Machine Learning and GeoAi (Python): TensorFlow, PyTorch, Scikit-Learn, YOLO, Leafmap, Geemap, Geopandas, Hypercoast, HyRiver, GEE Python API
Other Languages: C# (ASP.NET), R

Experience

Government of Pakistan, Geospatial Research Officer — Rawalpindi, Pakistan, May 2024 - Current

- Led in the development of state of the art web applications for geospatial data analysis and visualization
- Numerical modeling and simulation using multi-source satellite data for improved flood prediction, disaster risk assessment, and mitigation planning
- Processing of SAR, InSAR, and LiDAR data for advanced geospatial analysis.

Government of Pakistan, GIS Officer — Rawalpindi, Pakistan, Jan 2024 – May 2024

- Developed GIS-enabled dashboards and applications for visualizing geospatial analyses of floods, agriculture, and land management.
- Digitized and managed land information systems with datasets including land records,

crop yield, flood maps, electoral boundaries, and routing layers.
- Runoff estimation, watershed delineation, and numerical modeling for advanced spatial hydrological analysis.

National University of Sciences and Technology Islamabad, Research Associate — Rawalpindi, Pakistan, Jul 2023 – Dec 2023
- Satellite Mapping and Identification of Potential Antimony Deposits in the Pishin Basin, Balochistan, Pakistan
- Performed numerical modeling and simulations for coastal inundation and flood hazard evaluations.
- Generated reports and documentation for flood and water management projects, including pre- and post-flood analyses.
- Integrated multiple datasets in Google Earth Engine for advanced hydrological modeling and water quality monitoring.

Defense Housing Authority, Assistant Manager (GIS) — Rawalpindi, Pakistan, May 2022 – Jul 2023
- Led cadastral mapping projects, ensuring accuracy while managing a team to drive innovation, oversee operations, and align efforts with organizational goals.
- Monitored performance, provided feedback, and implemented solutions to enhance productivity and support land revenue projects, including future acquisitions.

Chinar Mines and Minerals, Snr GIS Analyst — Rawalpindi, Pakistan, Apr 2020 – May 2022
- Geological Mapping of potential mineral zones in different regions of Pakistan.
- Developed a GIS-enabled surveying web application for identifying and mapping Pakistan's mineral potential zones.
- Prepared the first integrated map of mineral potential zones in Pakistan, combining Earth observation data with field data and geochemical analysis.

Pakistan Council of Research in Water Resources, GIS Intern — Rawalpindi, Pakistan, Jan 2020 – Apr 2022
- Conducted water quality testing, surveying, and mapping for environmental assessments.
- Prepared comprehensive reports and documentation based on survey findings and analysis.

Projects

Projects Portfolio

- Explored and mapped placer gold and base metal deposits along the Hunza River, Gilgit Baltistan
- Identified minerals in Chitral KPK using hyperspectral remote sensing data, geological field exploration, and geochemical analysis
- Identification of Potential Antimony Deposits in the Pishin Basin, Balochistan, Pakistan using multi-source remote sensing data
- Assisted in end-to-end mineral exploration across Pakistan, conducting fieldwork, geochemical analysis, and remote sensing to identify chromite (Sapat Complex), polymetallic (Kargah Nala), coal (Duki, Sorrange, Lukhai Dadwaz), placer gold (Indus River), and mapped vein-type deposits (Reshun Fault), phosphate (Taranawai), and dolerite (Mansehra, Oghi), while supervising underground mining (Cu, Pb, Sb, Cr) with resource estimation and grade control
- Contributed to mineral exploration projects in Pakistan by integrating geochemical and remote sensing data for precision targeting of chromite, polymetallic, coal, and gold deposits, assisting in structural mapping and underground mining operations with rigorous resource evaluation.
- Modeling and Simulation of Badswat Glacier Debris Flow Hazard Assessment Using RAMMS Model
- Landslide identification, assessment and quantification in Azad Kashmir using re-

peatpass synthetic aperture radar (SAR) interferometry

- Simulating Coastal Inundation Vulnerability and Impact Assessment for the Indus River Delta
- Efficiently mapped flood hazards using drones and aerial imaging techniques: State of the Art and shortcomings
- Delimited electoral boundaries during the inter-census period using a hybrid approach integrating remote sensing and census data
- Developed a data management (CRUD) application using ASP.NET, Javascript, and Microsoft MySQL Database
- Created a Google Earth Engine application for land surface temperature calculation using Landsat series satellites
- Categorized soil based on its spatial variability for enhanced fertilizer application in Rawalpindi Tehsil, Pakistan
- Conducted pre- and post-flood monitoring of the 2022 Pakistan floods using Sentinel-1 SAR data
- Modeled the impact of urbanization on land surface temperature and assessed surface urban heat islands using Landsat: A case study of Rawalpindi Tehsil
- Developed a Web GIS application for mineral mapping in Chitral KPK using OpenLayers, PostgreSQL, and Geoserver
- Mapped flood susceptibility using GIS-AHP multi-criteria analysis (Nowshera, Pakistan)
- Modeled water quality parameters using Landsat 8 and ASD hyperspectral spectroradiometer data: A case study of Khanpur Dam
- Integrated multiple datasets in Google Earth Engine for advanced hydrological modeling using the Soil Conservation Service Curve Number method

Certifications

Certifications

- **Web GIS Development** (Udemy)
- **Spatial Data Science and Applications** – Yonsei University (Coursera)
- **GIS, Mapping, and Spatial Analysis Specialization** – University of Toronto (Coursera)
- **Spatial Data Science: The New Frontier in Analytics** – ESRI Training (MOOC)
- **Hyperspectral Thermal Image Unmixing** – Geo University
- **GIS for Climate Action (MOOC)** – ESRI
- **Imagery in Action (MOOC)** – ESRI
- **Cartography (MOOC)** – ESRI
- **Going Places with Spatial Analysis (MOOC)** – ESRI
- **Spatial Data Management with Google Earth Engine**
- **Big Data Analysis for Water-related Applications** - United Nations University
- **Introduction to Geospatial Data Analysis with ChatGPT and Google Earth Engine** - United Nations University
- **Active and Passive Satellite Data Analysis Using Cloud Computing for Surface Water/Flood Mapping** - United Nations University

Referees

Dr. Javed Iqbal
Dr. Ejaz Hussain
Dr. M Saleem Mughal
Dr. Ali Wahid

- Dr. Javed Iqbal (Professor)
- Dr. Ejaz Hussain (Professor)
- Dr. Muhammad Saleem Mughal (Assistant Professor)
- Dr. Ali Wahid (Assistant Professor)